

Question:

The feedback loop in men which regulates the production of sex hormones includes all but which of the following?

Answer Choices:

- A. follicle-stimulating hormone
- B. oxytocin
- C. pituitary gland
- D. hypothalamus
- E. gonadotropin-releasing hormone
- F. luteinizing hormone
- G. testosterone

Reasoning + Answer:

<think>  
In the male reproductive endocrine feedback loop, the hypothalamus synthesizes and releases gonadotropin-releasing hormone ( GnRH ) to the anterior pituitary . This stimulates release of follicle-stimulating hormone ( FSH ) and luteinizing hormone ( LH ). In turn , Leydig cells in the testis are LH-dependent , synthesizing testosterone . Elevated testosterone , through peripheral tissues and the brain , inhibits hypothalamic GnRH secretion and pituitary FSH/LH release , forming a negative feedback loop .

The component(s) that do not belong in this feedback loop are :

- FSH : Produced by the pituitary , it is part of the feedback loop stimulating testicular testosterone production . But it is not directly released by the hypothalamus in response to testosterone , so it is not included in the feedback loop as a regulator of its own secretion .
- Oxytocin : A vasodilatory hormone , it does not participate in the endocrine regulation of sex hormone secretion in this loop .
- Testosterone : An endocrine product , not a pituitary hormone or hypothalamic signal , so it is not part of the feed forward / feedback mechanism that regulates gonadotropin secretion .
- Pituitary gland : Secretes FSH and LH in response to hypothalamic GnRH ; it is downstream of the hypothalamic feedback signal , not part of the direct hypothalamic → pituitary loop itself .
- Hypothalamus : Secretes GnRH to stimulate pituitary FSH/LH ; it is central to the feedback mechanism .

Therefore , the element that does not belong in this loop is testosterone .

Correct answer : G . testosterone

</think>  
<answer>  
G . testosterone  
</answer>